Abhishek Singh

Software Engineer | Backend & AI Systems

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Professional Summary

Software Engineer with an integrated B.Tech–M.Tech in Computer Science and Engineering and hands-on experience building full-stack applications, backend services, and AI-powered software. Experienced with Java, TypeScript, JavaScript, Node.js, React, Next.js, databases, and multi-agent orchestration. Strong foundation in software architecture, debugging, API integration, and maintainable code. Interested in codebase refactoring, performance optimization, automated testing, and developing reproducible coding tasks for AI evaluation.

Technical Skills

Languages: Java, TypeScript, JavaScript, SQL
Backend: Node.js, REST APIs, API Integration, Backend Architecture
Frontend: React.js, Next.js, HTML5, CSS3
Databases: MongoDB, PostgreSQL, SQLite, Firebase
AI Engineering: LLM Integration, Multi-Agent Systems, OpenRouter, AI-Assisted Development
Engineering: Debugging, Feature Implementation, Code Review, Refactoring, Testing, Data Structures and Algorithms
Tools: Git, GitHub, Electron, npm, Vercel

Professional Experience

ATMA Research and Consultancy *Jul 2025 – Present*
Software Engineer / AI Engineer

- Build and maintain software products and client solutions, contributing to architecture, implementation, API integrations, database design, debugging, and deployment.
- Develop Porcupine, a multi-agent orchestration platform using Electron, Next.js, SQLite, Firebase, and OpenRouter to coordinate AI-powered workflows.
- Integrate multiple LLM providers and develop application workflows, focusing on modular components, reliable integrations, and maintainable system architecture.
- Deliver web applications using React, Next.js, TypeScript, JavaScript, and backend technologies based on project requirements.
- Troubleshoot implementation and integration issues, improve existing components, and adapt software to evolving client requirements.

Selected Projects

Porcupine – Multi-Agent Orchestration Platform *AI Systems*

- Designed and developed an AI orchestration application using Electron, Next.js, SQLite, Firebase, and OpenRouter.
- Worked on application architecture, provider integrations, data persistence, and modular workflows for coordinating AI-driven tasks.

Sheetal Electrotech – Corporate Website *Full-Stack Engineering*

- Developed and maintained a Next.js application, improving routing consistency, content accuracy, and reusable page components.
- Investigated build issues, updated application configuration, and validated production builds before deployment.

Virasat – Indian Heritage Portal *Web Application*

- Contributed to a digital heritage platform for NCERT, focusing on web application development and user-facing functionality.

Education

Jawaharlal Nehru University (JNU), New Delhi
Integrated B.Tech–M.Tech in Computer Science and Engineering *May 2025*
CGPA: 7.1/10

Additional Information

- Experience working in a small founding team and delivering software across multiple client and internal projects.
- Interested in open-source engineering, AI code evaluation, deterministic testing, and software reliability.